ELEG 3143 Assignment # 1

1. If A and B are subsets in the same space, find
   (a) $(A - B) \cap (B - A)$
   (b) $(A - B) \cap B^c$
   (c) $(A - B) \cup (A \cap B)$
2. A space $S = \{a, b, c, d, e, f\}$ has two subsets defined as $A = \{a, c, e\}$ and $B = \{c, d, e, f\}$. Find
   (a) $A - B$
   (b) $(B - A) \cup A$
3. State whether each of the following defined events is an elementary event.
   (a) Obtaining two heads when three coins are flipped
   (b) Obtaining an ace when a card is selected at random from a deck of cards
   (c) Obtaining three heads when three coins are flipped
4. If a piar of dice are rolled, determine the probability of each of the following events.
   (a) Obtaining a sum of 11
   (b) Obtaining a sum less than 5
   (c) Obtaining a sum that is an even number

5. A company manufactures small electric motors having horse power ratings of 0.1, 0.5, or 1.0 horsepower (HP) and designed for operating with 120 V single-phase (1-ϕ) AC, 240 V single-phase AC, or 240 V three-phase (3-ϕ) AC. The motor types can be distinguished only by their nameplates. A distributor has on hand 3000 motors in the quantities shown in the table below.

Horse Power	120 V 1-ϕ	240 V 1-ϕ	240V 3-ϕ
0.1	900	400	0
0.5	200	500	100
1.0	100	200	600

One motor is discovered without a name plate. Deteremine the probability of the following events.

   (a) The motor has a horsepower rating of 0.5 HP.

   (b) The motor is designed for 240 V single-phase operation.

   (c) The motor is 1.0 HP and is designed for 240 V three-pase operation.

   (d) The motor is 0.1 HP and is designed for 120 V operation.

6. A box contains 25 transistors, of which 4 are known to be bad. A transistor is selected at random and tested.

   (a) What is the probability that it is bad?

   (b) If the first transistor tests bad what is the probability that a second transistor selected at random will also be bad?

   (c) If the first transistor tested is good, what is the probability that the second transistor selected at random will be bad?